



Airport surface operations: A holistic framework for operations modeling and risk management



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ABSTRACT

Air transport is a key driver for social and economic development and its demand has increased steadily over the years. One crucial element of the air transport system is airports, and in particular, the airport surface, which facilitates the ground movements of aircraft and provides the link between ground and air. The nature of surface operations is such that it requires the input, coordination and cooperation of various actors. The complexity of these operations makes the system vulnerable, and therefore, the development and implementation of an effective Safety Management System are required. The current approach to surface safety management, however, is piecemeal and not integrated. Typically, a single occurrence type is investigated from the perspective of an individual stakeholder with the consequence that resulting proposals for safety mitigation measures are biased and limited in terms of their impact. This paper proposes a framework for a holistic risk assessment of airport surface operations that integrates the actions of all relevant stakeholders. Firstly, a process model of surface operations is developed using the conceptual framework of Business Process Modeling and the input of various data sources to achieve triangulation. Secondly, the causal factors underlying accidents and incidents are determined based upon a reference data set that combines 12 databases from airports, airlines, Air Navigation Service Providers, ground handling companies and regulators. The factors are summarized in a new taxonomy. Finally, a macroscopic scenario tool that supports the management of change, training and education, and safety communication functions of the SMS is introduced.

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1. Introduction

Air transport is a key driver for social and economic development and its demand has increased steadily over the years. Since the mid-1980s, passenger numbers have more than doubled and freight traffic has increased almost three-fold (ATAG, 2005). This trend is expected to continue over the next 20 years, with world passenger traffic (by revenue passenger-kilometers) expected to grow 5% annually and air cargo (by revenue tonne-kilometers) 5.2% per year (Boeing, 2012).

A key element of the air transport system is airports, whose main function is the provision of a safe and efficient transition of passengers and goods between the ground and airspace. The operations on the airport surface (i.e. runways and taxiways) are crucial to the achievement of this function. From an operational perspective, airport surface operations require the interaction of five main stakeholders (airport authority (i.e. airport operator), pilot, air traf-

fic control (ATC), ground handling, regulator) both to facilitate the ground movement of aircraft and vehicles, and to maintain the surface in a working condition. One key performance indicator (KPI) of such operations is safety, which can be defined as 'the state in which the possibility of harm to persons or the property damage is reduced to, and maintained at or below, an acceptable level through a continuous process of hazard identification and safety risk management' (ICAO, 2009). Safety is one of the International Civil Aviation Organization's (ICAO) strategic objectives to foster a global civil aviation system that consistently and uniformly operates at peak efficiency and provides optimum safety, security and sustainability (ICAO, 2012).

Because of the complexity of aircraft and related operations, the airport surface, however, has proven to be vulnerable and at risk of failure with the consequence that accidents and incidents may occur. A study by the Flight Safety Foundation (FSF) showed that from 1995 to 2008, a total of 1429 commercial transport aircraft were involved in aviation accidents. Approximately 30% of these accidents were runway related and led to 973 fatalities (FSF, 2009). In addition to accidents, incidents also occur on the airport surface and, therefore, the development and implementation of an effective Safety Management System (SMS) are required to ensure

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the highest level of safety for surface operations. In fact, airport surface safety has been acknowledged to be a key area of aviation safety by stakeholders worldwide (e.g. ICAO, 2010; NTSB, 2012).

The current way of addressing surface safety is characterized by an ad-hoc piecemeal approach. Previous initiatives to mitigate the risk of surface operations have addressed the topic from different viewpoints including regulatory bodies at a national and international level (e.g. ICAO, 2007), multinational aviation safety organizations (e.g. FSF, 2009), Air Navigation Service Providers (ANSPs) (e.g. FAA, 2010), aircraft manufacturers (e.g. Airbus, 2009), as well as action plans at a local airport level (e.g. training and awareness campaigns). These initiatives highlight both the divided attention directed towards surface safety and the biggest limitation in the safety management of surface operations: i.e. that current initiatives focus on single accident or incident (i.e. occurrences) types and take on the viewpoint of a specific aviation stakeholder. For instance, the Federal Aviation Administration's (FAA) 'Runway Safety Report' (FAA, 2010) contains the results of an analysis of incursions and their causal factors. It is based on an internal FAA database for incursions, with most occurrences reported by ATC. Although the current focused approach is understandable given the different responsibilities and interests of the involved stakeholders, it cannot be sufficient, as any safety mitigation strategy is biased through the viewpoint that is taken. Such an approach neglects the operations of the other relevant stakeholders, and the interactions and dependencies between them. Therefore, a holistic and robust safety assessment of airport surface operations that follows the requirements of SMS is needed.

This paper proposes a framework for a holistic risk assessment that integrates the actions of all relevant stakeholders in the context of surface safety. The framework comprises a process model of surface operations, the determination of causal factors underlying failure modes of these operations, and a macroscopic scenario tool to evaluate changes in the system architecture. The model is applicable to all relevant stakeholders in the context of surface safety and allows identifying the key drivers to surface safety. Based upon this effective safety mitigation strategies can be developed.

2. Background

The airport surface (i.e. manoeuvring area) is 'the part of an aerodrome to be used for take-off, landing and taxiing of aircraft, excluding aprons' (ICAO, 2004a). It is defined by its infrastructure and five main stakeholders that act and interact upon it (Fig. 1).

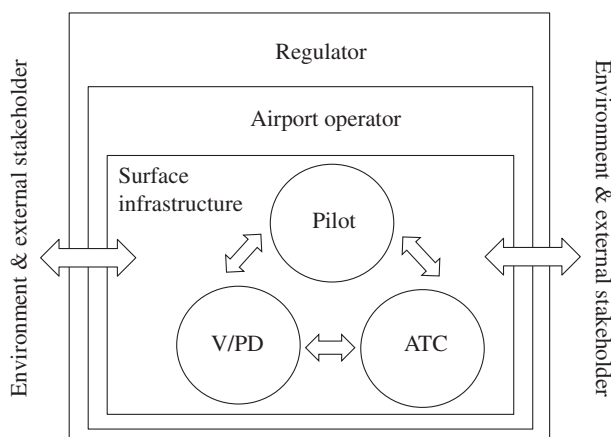


Fig. 1. Airport surface actors.

The manoeuvring area is used primarily by aircraft for landing, taxiing and take-off. Inbound flights land on a runway and then use the surface to taxi to their assigned ramp at the apron, where ground handling (i.e. servicing) takes place. Departure flights use the manoeuvring area to taxi from their ramp at the apron to an assigned runway for take-off. Pilots manoeuvre the aircraft. Besides aircraft, also vehicle drivers and pedestrians (V/PD) use the runway and taxiway system. The majority of vehicles and attended airport personnel are operating on the apron (e.g. ground handling vehicles and equipment). However, certain vehicles are allowed to enter the manoeuvring area. These vehicles include for instance tugs for towing aircraft, vehicles involved in maintenance and construction work, or snow removal equipment in wintertime. In addition, ATC plays a significant role for the safe flow of ground-movements. ATC's primary objective is the prevention of collisions between two aircraft in the air, or a collision between an aircraft and an obstacle on the ground, while expediting and maintaining an orderly flow of air traffic (EUROCONTROL, 2005). The airport authority is responsible for the management and oversight of an airport's operations. In terms of surface safety that means the management and maintenance of the airport surface, i.e. the provision of the infrastructure in working conditions. The airport authority operates on the airport surface in form of V/PD. Last, regulatory bodies oversee the system by providing a framework of rules and regulations and ensuring compliance of the different aviation stakeholders to those. In addition, the stakeholders are interacting with their environment.

The operations of the five stakeholders are dependent on each other. For instance, ground handling is responsible for servicing the aircraft (e.g. loading, fueling) while it is parked on the ramp. Only after the servicing is completed the aircraft can leave its parking position and start to taxi. To do so, the aircraft needs to be cleared from ATC. Thus, successful surface operations require the interaction of the various stakeholders. These interactions can be direct or indirect. For instance, pilots are in contact with ground handling over the dispatcher (US: 'The Lead'). Pilots are also in direct contact with ATC, e.g. to ask for pushback clearance. Ground handling, however, is linked only indirectly to ATC. In this example, the pilot needs to obtain pushback clearance from ATC and communicate it to the tug driver, who will eventually initiate the pushback.

Because of the complexity of the system, airport surface operations are vulnerable. To ensure that safety risks (including e.g. accidents/incidents) are identified, assessed and appropriately mitigated, aviation stakeholders are required to implement a Safety Management System (SMS). A SMS is a systematic approach to managing safety that is based on the four cornerstones of safety policy and objectives, risk management, assurance, and safety promotion. A SMS is a framework that provides an organization with the adequate tools to ensure that any drift by the organization towards a lower safety performance is prevented. At the heart of a SMS is the operational safety management (safety risk management), which supports the development of evidence-based measures for the overall safety management process. Safety risk management in practical terms is concerned with hazard and occurrence identification through reporting and data collection, investigation, and subsequent data analysis (ICAO, 2009). In particular, airport surface risk management is concerned with the collection, investigation, and analysis of four main accident/incident types: excursions, incursions/collisions, wildlife strikes, and Foreign Object Damage (FOD).

SMS is a systematic approach to safety and strives to assess and continuously improve the safety of an entire system. It therefore requires the assessment of all system components and their interactions for hazards and associated safety risks. Although airport surface safety has been addressed extensively in the past, previous research shows major limitations. To date no research exists that

analyses surface safety in an integrated manner. Safety mitigation strategies are developed for single occurrence types from the perspective of single aviation stakeholders reflecting only pieces of surface operations and associated failures/errors.

Typical examples for surface safety analysis and mitigation efforts include the European Organisation for the Safety of Air Navigation's (EUROCONTROL) Integrated Risk Picture (IRP) and the Causal Model for Air Transport Safety (CATS) developed by the Nationaal Lucht- en Ruimtevaartlaboratorium (NLR). IRP quantifies the risk for five accident scenarios to which Air Traffic Management (ATM) can make a significant contribution with the aim to identify safety priorities for ATM. Fault trees are developed for each accident scenario, including runway and taxiway collisions that are relevant for surface operations, and quantified using worldwide accident data. The risk of each specified accident category is quantified by providing a structured breakdown of their causes together with their probability of occurrence. In addition, common causal factors that affect all accident scenarios are separately presented in an influence model (EUROCONTROL Experimental Centre, 2006). Next to addressing only loss-of-separation-related accidents, IRP suffers from the reflection of a biased stakeholder perspective. Consequently, a large amount of information relevant to surface operations is not considered and, therefore, the model is of limited relevance to stakeholders other than ANSPs. In the context of runway and taxiway collisions this means that IRP does not model pilot or airport-related factors in detail and collisions with V/PD are missing entirely.

Similarly, CATS quantifies the risk of aviation accidents for 33 accident scenarios (Ale et al., 2009). While the model contains detailed event chains for excursions and incursions, these are limited to pilot and ATC factors and ignore the influence of such items as V/PD factors or airport design factors. The former account for 21% of runway incursions in the US (FAA, 2010) and 29% in Europe (EUROCONTROL, 2012). Similar concerns apply to the analysis of FOD and wildlife strikes, and it is apparent that CATS has been developed from a flight perspective rather than a broader ATM perspective.

The piecemeal approach implies that safety risk mitigation strategies will always be biased depending on the viewpoint of the stakeholder who initiated the control measures. Such bias might be acceptable at an organisational level as the stakeholders are interested in determining the risk related to their operations, however, it is also found at the highest level of legislation. For instance, the 'European Action Plan for the Prevention of Runway Incursions' (EUROCONTROL, 2011) depicts a consensus of subject matter expertise from primarily European ANSPs which provides recommendations for the mitigation of runway incursions applicable to all aviation stakeholders. The International Civil Aviation Organisation's (ICAO) 'Manual on the Prevention of Runway Incursions' (ICAO, 2007), which is based on the EUROCONTROL developments, has a similar intention and hence makes the recommendations in Europe effective on a global level. The ICAO's manual contains an initial reporting form for runway incursions. Through its biased development based on SME opinion, however, it fails to capture various factors that are critical in the context of runway incursions. For instance, the manual does not account for factors related to V/PD operations such as vehicle technical failures, V/PD competences, and team operations. A holistic approach as required in a SMS would allow understanding all the risks associated to surface operations.

In addition to a biased reflection of stakeholder viewpoints, previous analyses show methodological limitations. These include missing or ambiguous definitions, insufficient specification of safety data, and data quality concerns (e.g. data accuracy), and unspecified analysis methodologies. For instance, the FSF (2009) initiated a major study to analyse risk factors in runway excursions and recommended safety risk enhancements based upon the results of the analysis. The study is limited as it fails to take

undershoots, one type of excursion, as well as excursions from taxiways into account. It uses World Aircraft Accident Summary (WAAS) data, a worldwide database that aggregates detailed descriptions of accidents involving jet and turbo-powered aircraft, however, does not assess the quality of this data and its potential for aggregation. In addition, the method of extracting the risk factors from the safety data is not specified. Also the two models introduced before (IRP and CATS) suffer from the same limitations by aggregating data from different sources without quality checks. These limitations are consistent for industry initiatives (ATSB, 2008, 2009; FAA, 2010) and the academic literature (Es, 2005; Es et al., 2009; Valdes et al., 2011).

In summary, the current non-integrated approach to safety management has developed from the consideration of specific stakeholder interests. It neglects the relevant operations of the other relevant stakeholders, and their inter-dependencies, with the consequent impact of biasing resulting in less effective mitigation measures. Therefore, a holistic (integrated) safety assessment of airport surface operations is required, based on a reliable methodology and following the requirements (specified by the ICAO) of safety management systems.

This paper overcomes the gaps by proposing a framework for a holistic risk assessment of surface operations. The prerequisite for the development of a SMS is the description of a system architecture in terms of its components and their interactions. Such a baseline system can then be used to identify the safety vulnerability of human actors, other components and their interfaces (ICAO, 2009). This paper develops the baseline system of airport surface operations and subsequently assesses the system architecture in terms of safety risks. Firstly, a process model of surface operations is developed. Such a model helps understanding the tasks of the stakeholders and their interaction with physical airport surface infrastructure as well as the overall airport surface system architecture. Secondly, the causal factors to airport surface accidents and incidents are analyzed using a variety of research methods and the results are summarized in a new taxonomy. Thirdly, a qualitative model that shows how each component of the airport surface system architecture can fail is developed by mapping the causal factors to the process model. As the environment is dynamic and, therefore, characterized by change, such a model can also provide a macroscopic scenario tool to evaluate the impact of changes in the airport surface system and to guide further analysis. This paper is organized as follows. The next section introduces the methodology for the operations model and subsequent risk assessment. Part four presents the results and illustrates the benefits of a holistic approach, before the final part concludes the paper.

3. Methodology

The framework for a holistic risk assessment of airport surface operations was developed in four steps as shown in Fig. 2 and the following sections outline the methodology in detail. To begin with, a process model of airport surface operations was developed. Subsequently, the causal factors underlying airport surface accidents/incidents were analyzed and the results were used to refine the process model. Last, the operations model and causal factor analysis were merged to derive a macroscopic scenario tool. To achieve triangulation a variety of rigorous research methods were used in each of the steps.

3.1. Process model of airport surface operations

3.1.1. Theoretical framework and assumptions

First, a process model of normal airport surface operations was developed, i.e. the actions of the five main stakeholders (airport

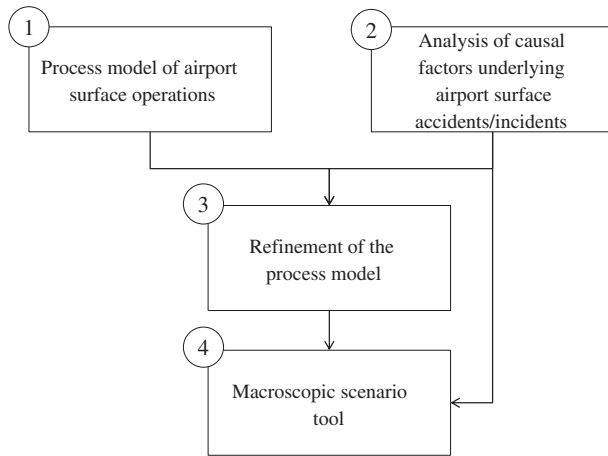


Fig. 2. High-level methodology.

authority, pilot, V/PD, ATC, regulator) and their interfaces were modeled in detail. The term ‘normal operations’ refers to airport surface operations in which the following assumptions are fulfilled:

- Aircraft, vehicles, and ATC equipment are not subject to malfunctions.
- The aeronautical infrastructure (including airport surface infrastructure) that is needed for normal operations is not subject to malfunctions.
- There are actors (i.e. pilot, V/PD, ATC) that interact for successful operations.
- The actors have the adequate level of training and are fit to carry out their work.
- Environmental conditions do not exceed the capabilities of the airport and its actors and,
- Airport operations are undertaken to an acceptable level of safety.

The process model was developed using the methodological framework of Business Process Modeling. Business Process Modeling is a management tool for improving organizational efficiency and quality. The central feature of the methodology is the Business Process Model (BPM), a graphical representation of the business process, also known as ‘notation’. In the broadest sense a BPM represents a sequence of activities underlying organizational processes. These activities can be modeled at different levels of detail and may include for instance actors, events, activities, associations, and document- and information flows, depending on the modeling objectives and model requirements. The aim of this methodology is to model the business processes of an organization, in order to analyze them, to identify deficiencies, and eventually to optimize them (Holt, 2009; White and Miers, 2008; Havey, 2005) as outlined in Fig. 3.

The name misleadingly suggests the application of Business Process Modeling for conventional business processes exclusively. However, Business Process Modeling can be applied to all types of organizations and industries, and in fact, has been successfully employed in aviation (e.g. airline business process models (Ploesser et al., 2009), passenger check-in models (Lincoln et al., 2010)). The Unified Modeling Language (UML) was used to represent the BPM graphically.

Unified Modeling Language (UML) is a visual modeling language that is used to create a model of a system. It has its roots in object-oriented software engineering and is mainly used for system (e.g. software) development. System development is a drawn-out process involving the definition of requirements, an in-depth anal-

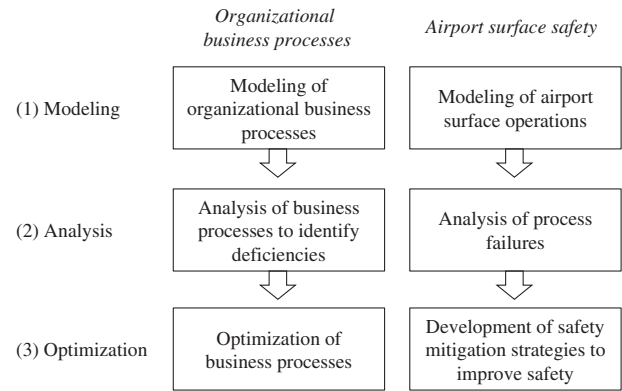


Fig. 3. Methodology business process modeling.

ysis of existing business processes, design, development and deployment. UML supports this process by providing a toolkit that allows the various elements of the current system to be visualised, the requirements for the system to be developed, and associations between the different elements to be made. Although UML was developed for the Information Technology (IT) industry, it has been applied to various areas, including the analysis of conventional business processes (Schmuller, 2009).

UML is not the only modeling language available, in fact, there are numerous languages available for process modeling. Whilst most of them represent processes as a series of steps, they emphasize different aspects of the process, such as activities, products, or document and information flows. Hence, they are suited for different kind of processes.

The aim is to describe the physical architecture of the airport surface as well as the operations of the stakeholders, i.e. the tasks each actor needs to carry out to complete his/her operation. Therefore, a modeling language that focuses on the flow of activities was required. Three examples often used in Business Process Modeling, which support activity modeling, are the Business Process Model and Notation (BPMN), UML, and the System Modeling Language (SysML). They have been developed for different applications and BPMN is mainly used for business analysis, UML for software development, and SysML for system engineering. Regardless their proposed applications, the execution of the activity diagrams in the notations is very similar and differs mainly in their terminology and graphics (Bock, 2006; IBM, 2013; Schmuller, 2009; White, 2013). Apart from small differences in the notations, the three languages are conceptually the same. Hence, all three modeling languages are suitable to model the flow of activities of the airport surface stakeholders. Also, research has shown that there is no difference in the usability of the three languages (i.e. the ease of use is no different) (Peixoto et al., 2013) and, therefore, the decision to use UML was made based on the availability of the required software.

3.1.2. Method for derivation of BPM

Individual models were developed for the physical airport surface infrastructure, the airport operator, pilot, V/PD, ATC, and the regulator and the interfaces between the individual parties were highlighted. The components were then integrated in a high-level architecture of the overall airport surface system. A high-level integrated architecture that specifies the components of a system is a necessary prerequisite for SMS development.

Firstly, the physical airport surface infrastructure was modeled using a class diagram in aggregation form. A class is a category of objects with similar attributes and aggregation diagrams are used to model hierarchies of classes. They show relationships in which one class is a component of another class (Schmuller, 2009).

In the second step, the role of the airport authority in managing and maintaining the physical surface infrastructure was defined. At this stage the responsibilities of the airport authority only are introduced, as airport authority personnel operate on the airport surface in form of V/PD. Details on how these responsibilities are executed are shown subsequently in a model of V/PD operations. The role of the airport authority was summarized in a package diagram, a type of diagram that is designed to group the elements of a diagram (Schmuller, 2009).

In a third step, the operations of pilots, V/PD, and ATC were modeled. To develop the notation Hierarchical Task Analysis (HTA) was applied. HTA, as specified e.g. in Kirwan and Ainsworth (1992), is a systematic method of describing how work is organized in order to meet the objective of the operation. It is a top-down approach, which starts with identifying the overall goal of the operation (objective). This operation is subsequently broken down into sub-operations (phases), which are accomplished through a series of tasks. In addition, the necessary resources for task accomplishment (i.e. actors that are required for task accomplishment) are defined. To achieve a detailed understanding of the tasks the stakeholders perform it is desirable to incorporate as many sources of information as possible, such as documentation, manuals for maintenance and operating procedures, and expert opinion from designers, operators, instructors and managers, as well as observations. When carried out correctly, HTA provides an exhaustive analysis of the problem addressed. This systematic deconstruction of the operation provides a picture of the tasks undertaken by each of the actors in normal conditions, and can be continued until the desired level of detail is reached (Kirwan and Ainsworth, 1992; Stanton et al., 2005). A further advantage of HTA is that it is a generic method that is adaptable to a wide range of purposes, and indeed, has been successfully applied in aviation research (e.g. Nascimento, 2011; Walker et al., 2005; Soekha, 1997).

To model pilot, V/PD, and ATC operations three separate models were necessary, as otherwise any model would always take on a specific viewpoint. A high-level task analysis was carried out for each of the three actors to decompose their operations into meaningful phases. After having decomposed the operations into phases, tasks, which need to be carried out in each of the phases, were defined. This research aims to be applicable to as many different aircraft, airport operators and airport operational personnel, and air traffic control units as possible. In order to keep any model in a manageable size the limits for task decomposition have been imposed in relation to:

- Aircraft model.
- Procedures at a company level (airline, airport, ANSP, ground handling companies), and
- Tasks of V/PD allowed on the airport surface.

Whenever further task decomposition would have required an analysis of sub-tasks, specifically applicable to a single aircraft model, a specific task of V/PD, or to attend the procedures of a given company decomposition should have stop. Subsequently, the actors that are required to accomplish the necessary tasks in each phase were identified. Two types of actors can be distinguished: direct actors and indirect actors. This distinction was derived from Reason's concept of active failures and latent conditions (Reason, 1997). Reason describes active failures as unsafe acts that 'are likely to have a direct impact on the safety of the system'. Active failures are errors or violations committed by people directly involved in operations (e.g. ATC, pilots). Latent conditions (e.g. poor design, undetected manufacturing defects, gaps in supervision) on the other hand are factors that lie dormant in the system may be present for

many years before they facilitate, or contribute to an occurrence (e.g. design failure).

The corresponding UML notation used state diagrams to model the task-deconstruction. State diagrams model the change an object goes through in response to events in time. The detailed task analysis for the pilots and V/PD was subsequently represented by activity diagrams that show the steps, i.e. activities, in an operation or process. Besides activities, the geographical location in which the tasks are accomplished were highlighted. The consideration will later help to identify how failures in the process of one actor's operation can affect the operations of the other stakeholders. Finally, the actors that influence the operations were summarized using a class diagram (Schmuller, 2009).

The UML notation for ATC operations is slightly different to that described above. Since ATC operations are multi-tasking operations, the activities cannot always be modeled in a pre-defined sequence. Activity diagrams showing the sequence of tasks seemed inappropriate and, consequently, a package diagram was employed. As above, however, the actors influencing ATC were summarized using a class diagram.

After having developed detailed operational models for the actors directly involved in surface operations (pilots, V/PD, ATC), the role of the regulator was defined in a fourth step. A package diagram summarized the regulator's responsibilities.

Fifth, the interfaces between the actors were highlighted using interface diagrams. Finally, based on the developed models, the high-level integrated architecture of surface operations, i.e. the *airport surface system*, was derived. This model presents all relevant stakeholders in the context of surface safety and their interactions and serves as baseline for SMS development.

3.1.3. Implementation of the method

HTA requires the input of various sources of information and, therefore, a hybrid methodology has been applied to develop the process model in four steps:

- (1) Review of the relevant literature and existing airport models (e.g. Dupuy, 2012; Subotic, 2007; Kazda and Caves, 2000; Ashford et al., 1997; Wells, 1996; Horonjeff and McKelvey, 1994), and
- (2) Analysis of operations manuals (e.g. easyJet, 2010a,b), standards and procedures (e.g. ICAO, 2006b,c, 2005a,b, 2004a,b, 2001a).

Based on the review of the literature and the analysis of operations manuals initial UML diagrams were developed for each stakeholder. These diagrams were subsequently refined using observations and structured communications with operational personnel as outlined below.

- (3) Observations and structured communications with relevant stakeholders:
 - Three pairs of pilots were observed during six jump seat flights (i.e. outbound and inbound flight per pair of pilots). The initial UML notation for pilot operations was used as a baseline and it was observed whether the pilots acted according to the model proposed. The first flight (i.e. outbound) was used each time purely to observe the pilots. During the second flight (i.e. inbound) the pilots were questioned as to when and why they did a certain task. In addition, the pilots explained more general issues (e.g. instruments and computer systems in the cockpit, documentation on board) that helped to clarify their operations and responsibilities. Based on these inputs the initial UML notation was refined (e.g. tasks were added, task sequence was changed).

- Likewise, pairs of ATC Tower controllers were observed at Gatwick Airport, Boston Logan International Airport and Oslo Lufthavn. The choice of these airports is explained in the next paragraph. Each position (i.e. planning and tactical controller) was observed for 15 min. The initial ATC UML diagram was used as a reference and it was observed whether the ATC operations conform to the proposed model. Subsequently, the tower supervisors at each airport explained in detail the tasks of ATC and the initial UML notation was refined using these inputs.
- Finally, observations at Heathrow Airport, Gatwick Airport, Boston Logan International Airport, Manchester Boston Regional Airport, Oslo Lufthavn and structured communication with airport authority operational personnel were conducted. These included airfield tours and structured communications with Airfield Superintendents (i.e. supervisors). The initial UML notation for the airport surface infrastructure guided the tours and the personnel were asked to show each of the components. In addition, other personnel operating on the airport surface were questioned on an ad-hoc basis about their tasks and responsibilities (e.g. construction workers, wildlife management personnel) in order to better ascertain the functionality of general airport operations. At Heathrow Airport the turn-around processes of several aircraft were observed and the Safety Manager of a ground handling company explained in detail the tasks of the various parties involved. Ad-hoc communications with operational personnel on the apron helped further to understand the role of ground handling. The observations spanned over a whole day at each airport in order to cover both, peak and non-peak, hours and the initial UML diagrams were refined.
- The airports used as inputs for the model development were chosen to fulfill the following criteria:
 - a. Mixture of airports in North America and Europe to cover operational differences, and
 - b. Medium to large airports in terms of movements and passenger boarding to ensure that a sufficient number of operations take place during the observation period (e.g. small airports may only have very few and isolated operations per day).

Following the model development the UML notation was validated internally (i.e. with operational personnel from airports that had been used as input for the model development) and externally (i.e. with experts from new airports and organizations).

- (4) Validation of the developed models with subject matter experts (SMEs):
- Shift supervisors of a major European and US airport authority,
 - FAA tower supervisor and tower supervisor at a European airport,
 - Chief Pilot of a US-based airline and a Safety Manager of a European airline,
 - FAA and EUROCONTROL.

All SMEs had at least fifteen years of relevant experience in their particular area and were in senior positions. As of November 2005, airports are required through Annex 14 on the Convention on International Civil Aviation (ICAO, 2004a) to have a SMS in operation. However, a safety-critical thinking has emerged in the aviation industry already in the nineties (Stolzer et al., 2011) and, therefore, a minimum of 15 years of aviation experience ensured

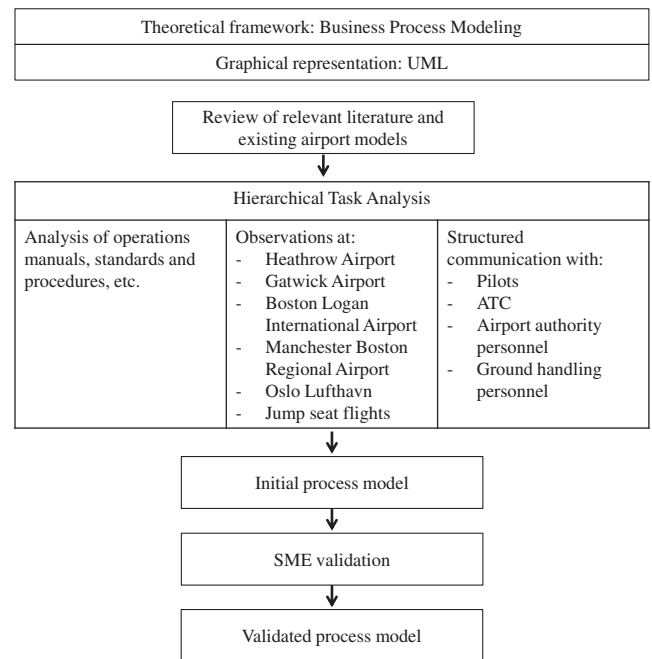


Fig. 4. Methodology for the development of an initial process model.

both, knowledge about operations and the necessary awareness for safety. The relevant parts of the UML were presented to the SMEs and they were asked to comment on it.

Fig. 4 summarizes the methodology for the development of the initial process model.

3.2. Analysis of causal factors underlying airport surface accidents/incidents

3.2.1. Method for identification of causal factors

After the airport surface system has been modeled, the safety of the system and its drivers were analyzed. Measures of safety are accidents and incidents. In order to understand as to why airport surface operations can become unsafe and lead to occurrences, the underlying causal factors to accidents/incidents needed to be understood in detail.

To analyze the causal factors underlying airport surface occurrences template analysis was chosen as the most appropriate method. This approach for qualitative data analysis involves the use of an initial template, which is refined over time. Such a template can be developed based on the literature, personal experience, case studies and subject matter interviews, for example, and the initial template is refined by the incorporation of new material. New codes can be added if a subject is not captured in the template, codes that are not used can be deleted, and codes that have been initially defined at a lower level can be put at a higher level and vice versa (Symon and Cassell, 1998). Furthermore, structured communications with SMEs in senior positions in the area of airport surface safety at the FAA and EUROCONTROL revealed that approaches such as grounded theory, whereby a theory is developed newly from data (Glaser and Strauss, 2009), are not advisable for this research as many factors or groups of factors are already known. It was advised to use the existing state of knowledge and if necessary to refine it.

To overcome the limitations highlighted in the literature, this research used a hybrid method that combined four research methods to analyze the causal factors. The approach integrated literature findings with a multi-national analysis of safety data, interviews with SMEs, and an airports survey.

3.2.2. Implementation of the method

An initial template of causal factors was developed based on the review of the literature on airport surface safety (e.g. [FAA, 2010](#); [FSF, 2009](#); [UK CAA, 2008](#); [ICAO, 2007](#)), accounting for 344 factors for all surface safety occurrence types.

The initial template was refined by the results of the analysis of safety data to inject a sense of realism based on actual operational experience and provision of a historic perspective of airport surface safety occurrences and their underlying causal factors. Safety data of ten organizations in four countries (United States, United Kingdom, Norway, and New Zealand) were collected. Requirements for data collection have been derived based on the limitations of previous research on airport surface safety as outlined elsewhere ([Wilke and Majumdar, 2012](#); [Wilke et al., 2012a](#)). Particular care was taken as to include only countries known for a good safety culture (in North America, Europe, Oceania) to ensure a high reliability, continuity, and completeness of the data. The data sets covered the viewpoints of airport authorities, airlines, ANSPs, ground handling companies, and regulators. [Table 1](#) provides an overview on the collected data, its geographical distribution and the source aviation stakeholder.

The data captured all airport surface accident and incident types (i.e. excursions, incursions/collisions, wildlife strikes, FOD). To refine the initial template, the descriptive narrative of each incident was analyzed and a code assigned. In addition, new codes were added to the initial template. A total of 6300 occurrences were analyzed. To validate the data coding, Cohen's Kappa inter-rater reliability test was applied, which computes the extent of agreement between the two raters, taking into account the agreement occurring by chance. Another qualified rater was asked to code 25% of the data, randomly selected. The coding can be accepted as reliable if the agreement rate is at least 70 ([Fleiss et al., 2003](#)). To account for differences in data quality, the data sets were analyzed for internal and external validity, as well the reporting scope of the underlying reporting system ([Wilke et al., 2012b](#)).

Following the data analyses, SMEs were interviewed in order to complement the data analysis, to gain detailed insight into the causal factors, and to enhance the understanding of certain fuzzy areas. For instance, it was not possible to collect a data set from a US-based airline due to confidentiality reasons. In order to still capture this perspective, interviews with 20 US-pilots were conducted.

Last, an airports survey with 58 airports in North America, Europe, and Oceania analyzed the impact of airport characteristics

Table 1
Data overview.

Organization	Stakeholder	Geographical distribution
UK Civil Aviation Authority (UK CAA)	Regulator	United Kingdom (UK)
Civil Aviation Authority of New Zealand (CAA NZ)	Regulator	New Zealand (NZ)
Civil Aviation Authority – Norway (CAA-Norway)	Regulator	Norway
Federal Aviation Administration (FAA)	Regulator & ANSP	United States (US)
Signature Flight Support (Signature)	Ground Handling	United States (US)
Avinor AS (Avinor ANSP)	ANSP Division	Norway
Avinor AS (Avinor Airports)	Airports	Norway
Oslo Lufthavn AS (OSL)	Division Airport operator	Norway
Norwegian Air Shuttle ASA (Norwegian)	Airline	Mainly Europe
Widerøe Flyveselskap AS (Widerøe)	Airline	Mainly Scandinavia
easyJet plc (easyjet)	Airline	Mainly Europe

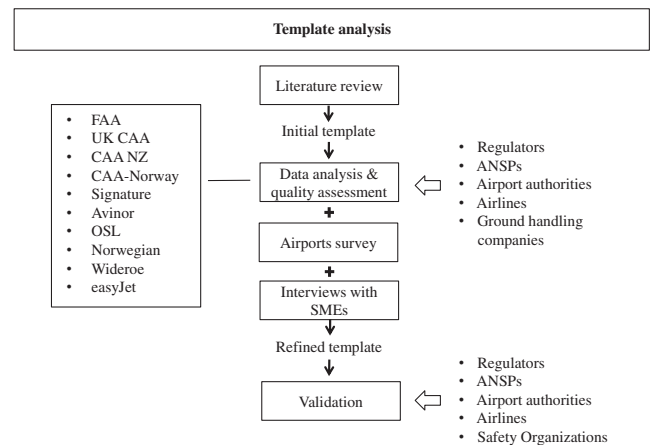


Fig. 5. Methodology for the analysis of causal factors underlying airport surface safety occurrences.

(e.g. manoeuvring area layout) on the occurrence of accidents/incidents. The results are discussed in detail elsewhere ([Wilke and Majumdar, 2012](#); [Wilke et al., 2012a](#)).

The final template of causal factors was validated with SMEs as outlined in Section 3.1.3.4. [Fig. 5](#) summarizes the methodology.

3.3. Model refinement

In a third step, the results from the causal factor analysis were used to identify random elements in the surface operations. For instance, links that do not exist in theory, however, have been identified through data analysis, were added to the model. The results from step 2 were used to refine the model of the airport surface system.

3.4. Macroscopic scenario tool

Last, the process model of airport surface operations and the results from the causal factor analysis were matched creating a qualitative model of airport surface operations that allows identifying the key drivers to surface safety. The final model shows all stakeholders involved in surface operations, their interactions, and matches the factors underlying process failures leading to accidents/incidents to each stakeholder's operations.

4. Results and discussion

4.1. Process model of airport surface operations

4.1.1. Physical airport surface infrastructure

The physical airport surface infrastructure comprises the runways and taxiways and associated safety areas, other surface areas (e.g. grassland between the surfaces), electronic and visual navigational aids (NAVAIDS) that assist aircraft in determining their position, as well as data collection devices (e.g. weather), radio transmission devices and other surface infrastructure (e.g. ground radar). The physical infrastructure has been modeled as a class diagram in aggregation form and is shown in [Fig. 6](#).

4.1.2. Responsibilities of the airport authority

The airport authority's role is the management and maintenance of the airport surface, i.e. the provision of the infrastructure in a working condition. The airport authority's operations therefore include the inspection and maintenance of the manoeuvring area (e.g. foreign object debris inspections, runway/taxiway pavement

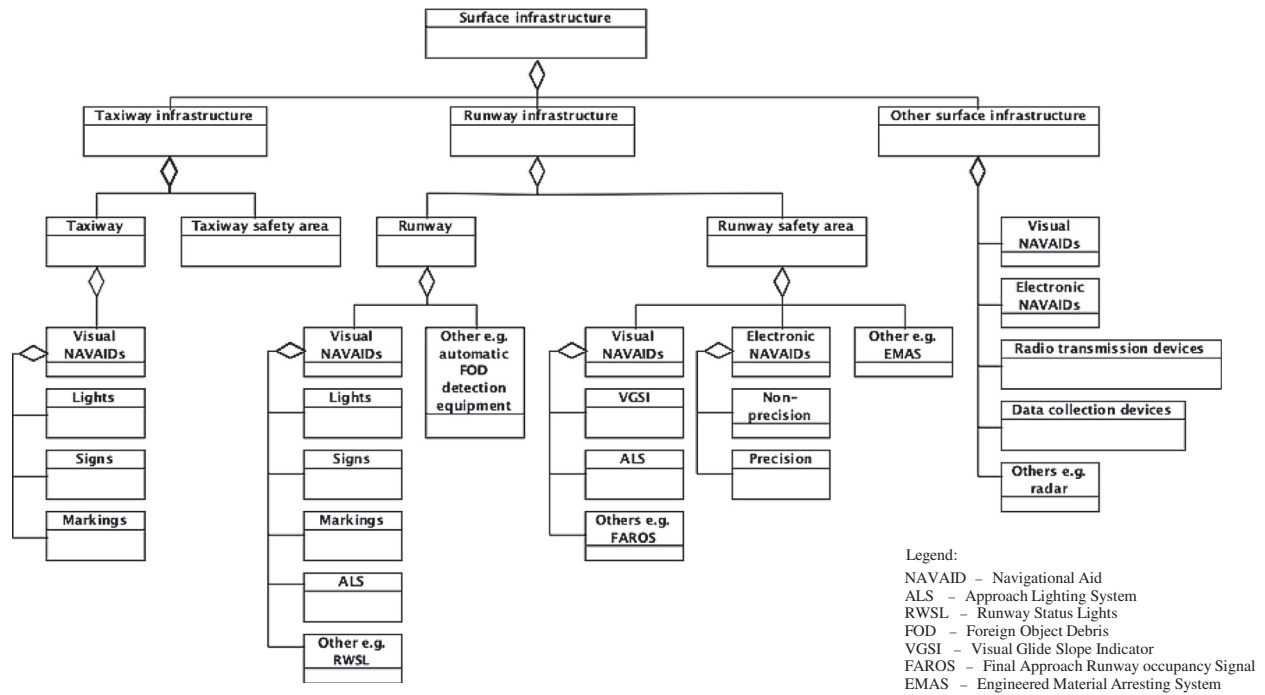


Fig. 6. Airport surface infrastructure.

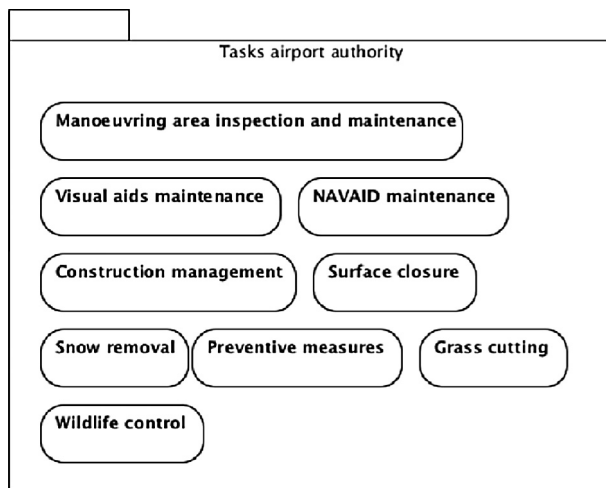


Fig. 7. Tasks of the airport authority.

maintenance) and the maintenance of the visual and electronic NAVAIDs. In addition, it is the airport operator's responsibility to close surfaces, for instance, during maintenance and construction periods. Construction management that includes the control of

works and site management are also part of the authority's operations, as well as grass cutting, snow removal, wildlife control, and the initiation of preventive measures (e.g. spraying for bugs, wheat). In UML notation the airport authority's tasks can be summarized using a package diagram as shown in Fig. 7. The airport authority operates on the surface in form of V/PD and details on how these operations are conducted (i.e. task analysis) are described in Section 4.1.4.

4.1.3. Pilot model

The objective of a pilot's operation is the safe and efficient conduction of flights. A pilot's operation (i.e. flight operation) can be decomposed into nine flight phases, as derived from ICAO (2006a) and shown in Fig. 8.

In the pre-flight phase the flight (i.e. pilots) and cabin crew board the aircraft with the intention to fly. The flight crew accepts the aircraft as airworthy and conducts the necessary pre-flight preparations. Subsequently, the aircraft is manoeuvred from the apron/ramp to its assigned runway, where it departs its origin and becomes airborne. The aircraft ascends in the Terminal Manoeuvring Area (TMA) until it reached its flight level and then cruises en-route. Once the destination is reached, the aircraft leaves its flight level and starts its descent. After touchdown at its destination airport, the pilots manoeuvre the aircraft from the

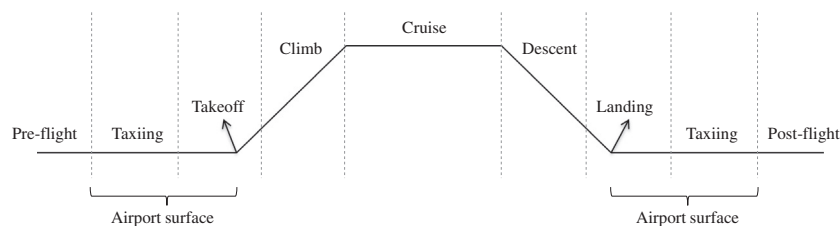


Fig. 8. Flight phases.

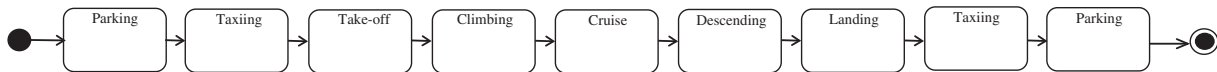


Fig. 9. Aircraft state diagram.

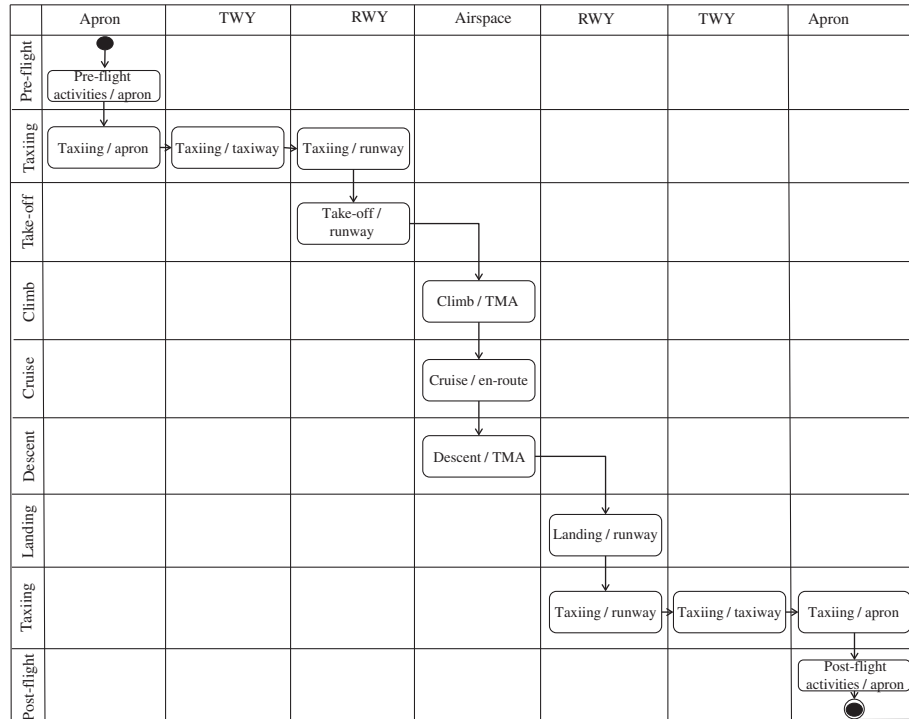


Fig. 10. Outline activity model – pilot.

runway to its assigned ramp at the apron and carry out the necessary post-flight inspections, before leaving the aircraft. The corresponding UML notation is shown in Fig. 9. The state diagram highlights the state of the aircraft during the different flight phases.

The nine flight phases can be further deconstructed into tasks. To present the model of normal pilot operations in a more concise manner, only a high-level activity diagram is provided. Fig. 10 indicates the tasks to be carried out in each phase of flight. In addition, spatial effects are shown, i.e. the location on the airport surface and the airspace, in which the tasks are carried out. Table 2 subsequently sets out a detailed outline of pilot operations, indicating the phase of flight and the tasks to be performed during each phase. In addition, the physical location of the aircraft during the flight operation, as well as spatial transition points (e.g. transition from taxiway to runway) are highlighted. The spatial analysis will help later on to define the interfaces between the different stakeholders.

Fig. 11 defines the actors directly and indirectly involved in the safe and efficient conduction of flight.

The pilot is in direct contact with the cabin crew, the engineer responsible for aircraft maintenance, and the dispatcher (US: The Lead) responsible for loading and unloading the aircraft. The dispatcher again is the connector between the pilot and ground handling personnel, as well as the gate agent (i.e. personnel responsible for boarding operations). In addition, the pilot is in contact with ATC and V/PD (e.g. operational personnel that is connected to the pilot on the radio initiating the pushback). On the other hand, pilots are indirectly influenced by the airline, aircraft

manufacturers and maintenance engineers, other V/PDs (e.g. ground handling personnel) and customers.

4.1.4. V/PD model

The overall objective of V/PDs is the safe and efficient conduction of a specific task. The operational phases were defined on a high level so as to make the model fit to any V/PD operation on the manoeuvring area. V/PD operations can be summarized into five distinct phases: pre-task, taxi-in, task accomplishment, taxi-in, and post-tasks, as outlined in Fig. 12.

The vehicle driver boards the vehicle with the intention to carry out a task. If applicable, he/she also carries out the necessary preparations for the operation, for instance, getting the snow removal equipment ready. Subsequently, the vehicle moves from its parking position (point A) to a point B on the airport surface, where it accomplishes its task (e.g. snow removal, runway inspection). Some task may require the parking of the vehicle on the manoeuvring, for instance, when the driver leaves the vehicle to inspect the infrastructure. Once the task has been carried out the driver manoeuvres the vehicle from point B back into its parking position. If required, post-operation tasks are carried out (e.g. getting the snow removal equipment ready for its next use), before the driver leaves the vehicle. The corresponding UML notation, shown in Fig. 13 highlights the state of a vehicle during the different phases of operation.

The five phases of V/PD operations may be further deconstructed. Fig. 14 indicates the activities for each phase using a high-level activity diagram. Spatial effects are considered and

Table 2
Task analysis – pilot model.

Flight phase	Tasks to be performed	Spatial transition points
Pre-flight	Flight preparation (e.g. weather briefing, Notice to Airman (NOTAM) briefing (e.g. RWY/TWY closure), cabin crew briefing) Check of aircraft maintenance records and acceptance of the aircraft as airworthy Cockpit preparation (e.g. FMGS data insertion, calculations) Exterior inspection (walk-around) Take-off briefing (commander/first officer) Weight and balance calculations Before start checklist Communication with ATC for clearance to push-back and start engine Check area clear of obstacles and other traffic Pushback Engine start After start checklist	Apron
Taxiing	Communication with ATC for clearance to taxi and assignment of the next stopping point (can be an iterative process) Check area clear of obstacles and other traffic (can be an iterative process) Command aircraft movement (can be an iterative process) Before take-off checklist Monitor clearance from obstacles and other traffic ^a Communication with ATC for clearance to enter the runway Check area clear of obstacles and other traffic Enter runway	Apron -> TWY TWY -> RWY
Take-off	Communicate with ATC for clearance to take-off Finish before take-off checklist Check area clear of obstacles and other traffic Take-off Monitor power margins and instruments, monitor speed increase, adjust flight profile accordingly ^a Monitor clearance from obstacles and other traffic ^a	 RWY -> TMA
Climb	Communicate with ATC for climb profile instructions (can be an iterative process) Adjust speed as required, monitor power margins (can be an iterative process) Gear up Flaps up After take-off checklist (~1000–1500 ft.) Decide when to turn on the autopilot (depending on personal preference from 400 ft. to 10,000 ft.) Monitor flight profile execution as instructed, adjust flight profile accordingly ^a Monitor clearance from obstacles and other traffic ^a	
Cruise	Communicate with ATC for cruise flight profile instructions (can be an iterative process) Adjust flight profile and instruments accordingly (e.g. altitude, speed, heading) (can be an iterative process) Monitor clearance from obstacles and other traffic ^a Cruise checklist or cruise briefing Briefing (for approach/landing)	TMA -> en-route
Descent	Communicate with ATC for clearance to initiate decent Monitor flight profile execution as instructed, adjust flight profile accordingly ^a Monitor clearance from obstacles and other traffic ^a Approach checklist (~18,000 ft., ~10 min before landing) Landing checklist (to be completed within the last 2000 ft.) Stable approach (for IFR 1000 ft., for VFR 500 ft.) Disable autopilot, assume manual control of the aircraft	En-route -> TMA
Landing	Landing Leave runway After landing actions After landing checklist	TMA -> RWY RWY -> TWY
Taxiing	Communicate with ATC for clearance to taxi and assignment of the next stopping point (can be an iterative process) Check area clear of obstacles and other traffic (can be an iterative process) Command aircraft movement (can be an iterative process) Shut engines down (e.g. aircraft with 2 engines shuts one down) Monitor clearance from obstacles and other traffic	 TWY -> Apron
Post-flight	Parking Engines – off Parking checklist Securing the aircraft checklist	 Apron

^a Monitoring is a permanent process.

a detailed outline of V/PD tasks is given subsequently in Table 3.

Fig. 15 provides an overview on the actors involved in V/PD operations (see Table 3).

4.1.5. ATC model

As mentioned before, ATC's primary objective is the prevention of collisions between two aircraft in the air, or a collision between

an aircraft and an obstacle on the ground, while expediting and maintaining an orderly flow of air traffic (EUROCONTROL, 2005). When mapping ATC's services to the flight phases introduced in Section 4.1.3 seven ATC phases could be derived. The phases have been derived from the subject matter literature (e.g. Cook, 2007; Subotic, 2007; ICAO, 2001b) and are outlined in Fig. 16.

Air traffic control operations are multi-tasking operations. A controller can handle multiple aircraft and, or vehicles at a time.

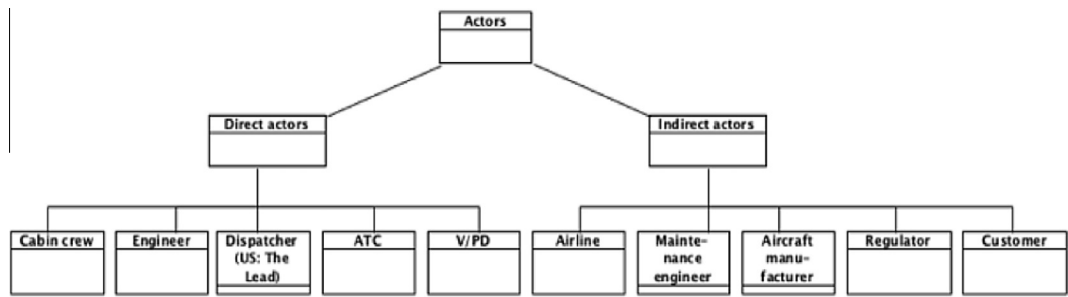


Fig. 11. Actors in contact with pilots.

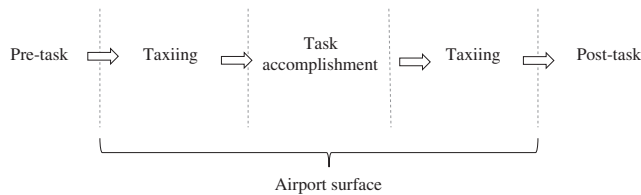


Fig. 12. V/PD operational phases.

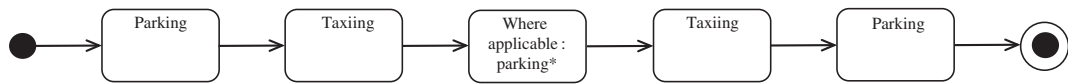
Although tasks can be defined for ATC operations, no order can be assigned, as several tasks can be accomplished at the same time. For instance, ATC can monitor ground movements, while talking to an aircraft over radio. When decomposing ATC's

operations further, four tasks and three sub-tasks were identified as follows:

- Traffic planning (traffic sequence plan),
- Execution of traffic sequence plan,
- Traffic adjustment through: identification of the adjustment need, adjustment plan, and plan execution, and
- Provision of separation through: conflict detection, conflict resolution plan, and traffic adjustment according to the resolution plan.

Task achievement through operational sub-tasks:

- Communication (real-time information and data),
- Navigation through instructions and clearances to aircraft and V/PD, and
- Surveillance (traffic monitoring).



* Specific tasks may require the parking of the vehicle during task accomplishment.

Fig. 13. V/PD state diagram.

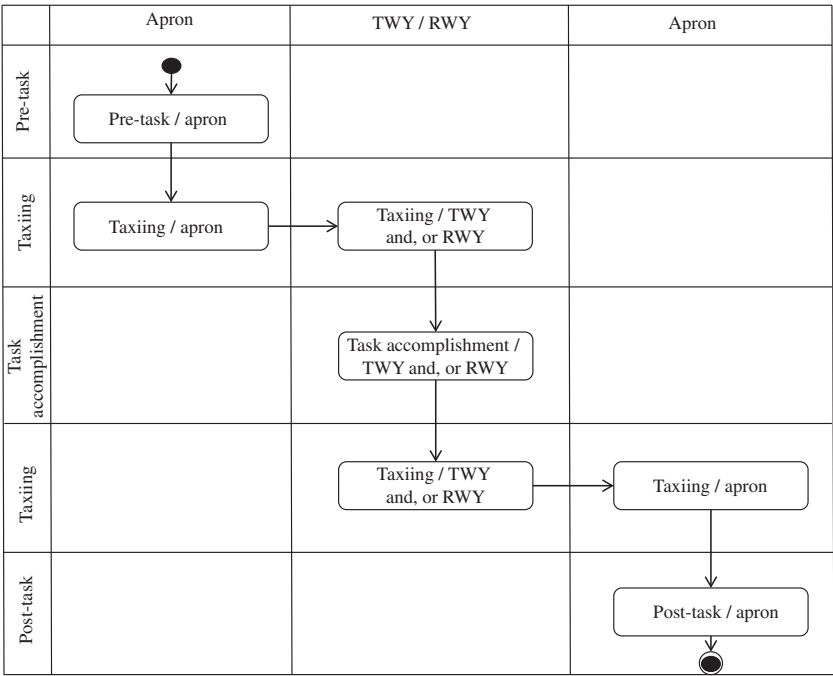


Fig. 14. Outline activity model – V/PD.

Table 3
Task analysis – V/PD model.

Flight phase	Tasks to be performed	Spatial transition points
Pre-task	If applicable, execution of necessary preparations for task accomplishment (e.g. get snow removal equipment ready) Command surface movement on the apron Communication with ATC to get clearance to enter the manoeuvring area Command surface movement	Apron Apron -> TWY
Taxiing	Communication with ATC for clearance to taxi and assignment of the next stopping point Check area clear of obstacles and other traffic Command surface movement Monitor clearance from obstacles and other traffic ^a Report clear once the requested runway/taxiway has been cleared [Taxi operations can be an iterative process]	(Depending on the tasks: a transition from TWY -> RWY can take place)
Task accomplishment	Accomplish task (e.g. runway inspection) Communication with ATC for surface movement instructions necessary for task accomplishment and adjust movements accordingly Monitor clearance from obstacles and other traffic [Communication with ATC (i.e. request for clearance, clearance confirmation, surface movement, report clear of runway/taxiway) can be an iterative process e.g. for each runway entry during the inspection]	
Taxiing	Communication with ATC for clearance to taxi and assignment of the next stopping point Check area clear of obstacles and other traffic Command surface movement Monitor clearance from obstacles and other traffic ^a [Taxi operations can be in iterative process] Report clear of manoeuvring area	(Depending on the tasks: a transition from RWY -> TWY can take place) TWY -> Apron
Post-task	Parking If applicable, execution of necessary post-task-processing	Apron

^a Monitoring is a permanent process.

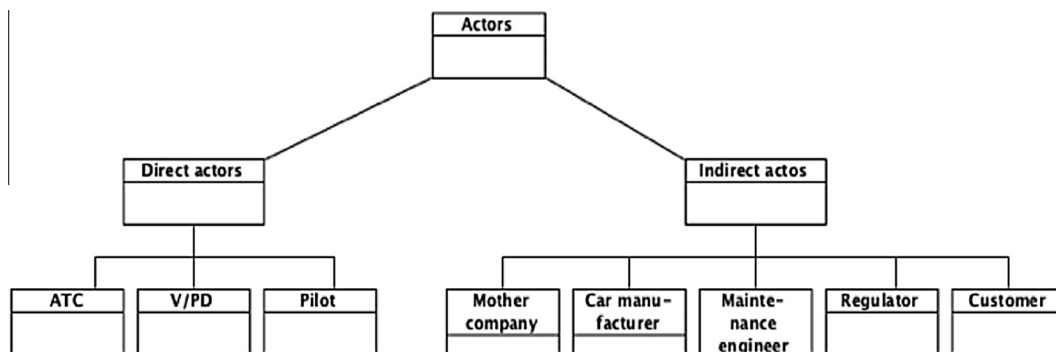


Fig. 15. Actors in contact with V/PD.

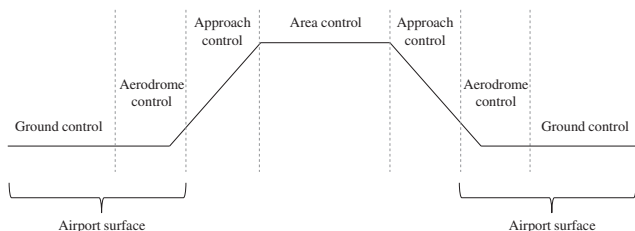


Fig. 16. ATC operational phases.

The tasks are the same for each ATC type (i.e. ground control, aerodrome control, approach control, area control) and they can be summarized using a package diagram as shown in Fig. 17. Direct

and indirect actors contribute to the successful operation of ATC (Fig. 18).

4.1.6. Responsibilities of the regulator

ICAO requires its member states 'to provide for the safe and efficient operations of aircraft within its airspace' (ICAO, 2006c). Each state that signed the Chicago Convention is obligated to provide e.g. adequate aerodromes, navigation aids, charting and instrument approach minima, weather reporting, air traffic services, search and rescue, aviation security, and timely correction of safety deficiencies with respect to these obligations. A Civil Aviation Authority (CAA) shall execute the proactive supervision and regulation of aviation activities. Under the convention and its annexes, it is the states responsibility to provide, and hence the CAA's responsibility to execute the tasks shown in Fig. 19.

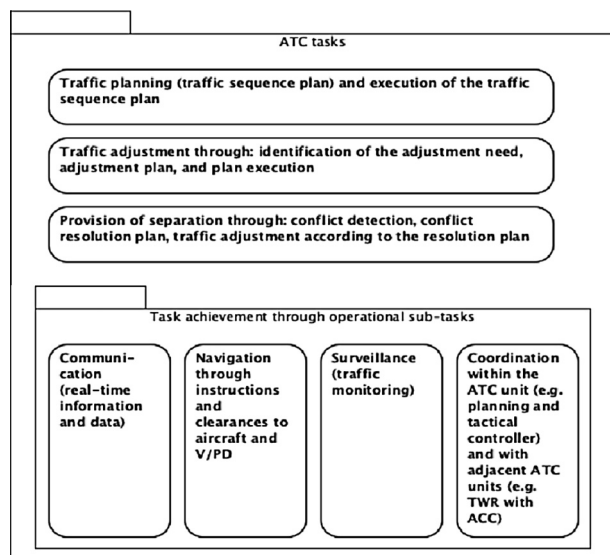


Fig. 17. ATC operations.

4.1.7. Interfaces

The previous sections introduced the operations and tasks for each of the main actors in isolation. In reality, these stakeholders interact. The interface between the three stakeholders that interact directly on the airport surface (i.e. pilot, ATC, V/PD) is communication (i.e. information and data) and shown in Fig. 20.

4.1.8. The airport surface system

Bases on the developed models for each of the aviation stakeholders (Sections 4.1.1–4.1.7) the *airport surface system* was derived. This model presents the high-level architecture of the system's components and shows all relevant stakeholders in the context of surface safety and their interactions. The airport surface system is shown in black font. Additions, as outlined in the next two sections, are added in color.

4.2. Taxonomy of causal factors underlying airport surface accidents/incidents

The outcome of the analysis of causal factors underlying airport surface occurrences is a new taxonomy and the details of its development are outlined elsewhere (Wilke and Majumdar, 2012; Wilke et al., 2012a). The taxonomy captures aircraft/pilot, ATC, airport/V/PD, environmental and regulatory factors. It is constructed in a hierarchical manner and captures a total of 736 factors on the lowest level of eight levels. The first three levels are constructed as follows:

1 Aircraft operations

11 Aircraft – technical

- 111 Aircraft technical failure
- 112 Aircraft weight and balance
- 113 Master caution warning
- 114 Aircraft design/manufacturing
- 115 Aircraft improper maintenance
- 116 Environmental influences
- 117 Other

12 Pilot – human

- 121 Resources
- 122 Competence
- 123 Communications
- 124 Situational awareness
- 125 Human-machine interface
- 126 Human reliability

127 Team operations

128 Procedures

2 ATC

21 ATC – Technical

- 211 Surveillance system
- 212 Navigation system
- 213 Communication system
- 214 Equipment design/manufacturing
- 215 Equipment improper maintenance
- 216 Other

22 ATC – Human

- 221 Resources
- 222 Competences
- 223 Communications
- 224 Situational awareness
- 225 Human-machine interface
- 226 Human reliability
- 227 Team operations

228 Procedures

3 Airport operations

31 Airport – technical

- 311 Vehicle technical failure
- 312 Radio communications
- 313 Ground handling equipment (cargo dollies, baggage carts, etc.)
- 314 Tugs

315 Airport meteorological equipment

316 Vehicle/equipment design

317 Vehicle/equipment improper maintenance

318 Other

32 V/PD – human

- 321 Resources
- 322 Competences
- 323 Communications
- 324 Situational awareness
- 325 Human-machine interface
- 326 Human reliability

327 Team operations

328 Procedures

329 Airport management

3210 Airport operations

33 Airport Physical

331 Design

332 Surface infrastructure and conditions

4 Environment

41 Weather

42 Geological disturbances

43 Night time operations

44 Wildlife

45 External factors

5 Regulatory system

51 Regulations not existent

52 Regulations inadequate

53 Regulations existent, but not implemented

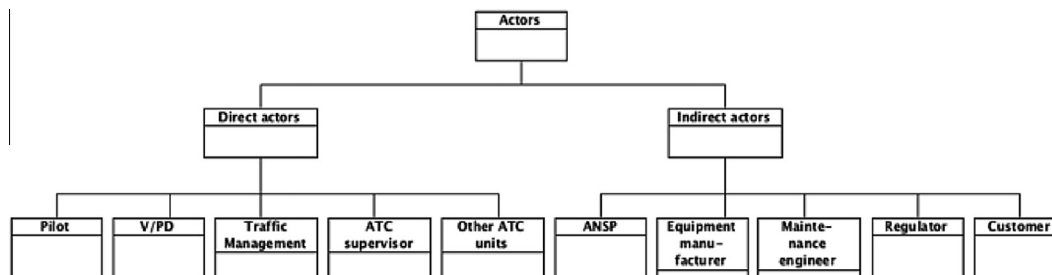


Fig. 18. Actors in contact with ATC.

The taxonomy provides a common industry language of the causal factors and a state-of-the-art tool that supports the data analysis function in the context of SMS. Through its robust

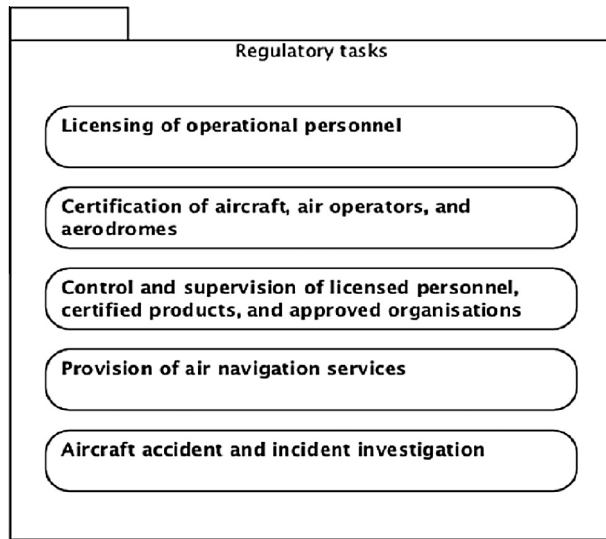


Fig. 19. Tasks of the regulator.

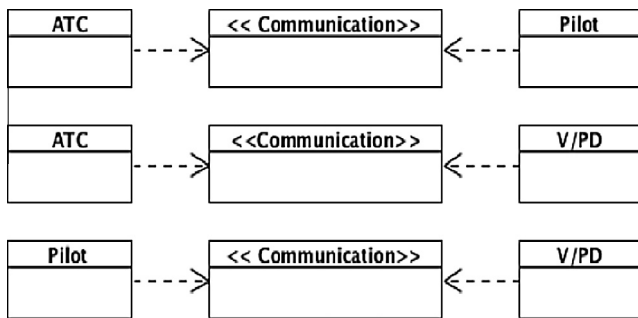


Fig. 20. Interfaces.

development the new taxonomy fits airport surface safety occurrences better than existing classification schemes (Wilke and Majumdar, 2012). Its unique feature is its integration and

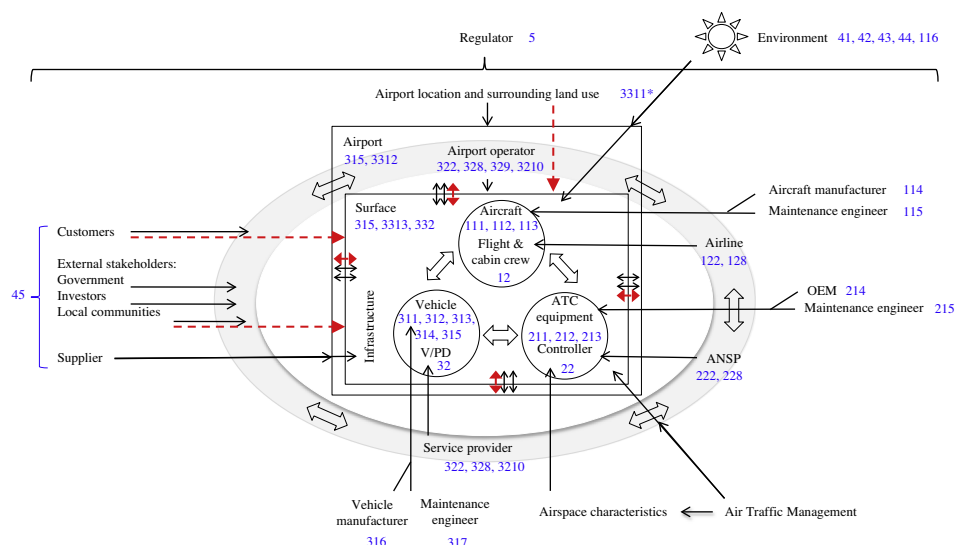
applicability to all relevant aviation stakeholders in the context of surface safety, and therefore, it is recommended for the operational safety risk management of all aviation stakeholders.

4.3. Random elements

After the causal factors underlying airport surface accidents/incidents have been analyzed, the results were used to identify random elements in the process model and to refine the model of the airport surface system. Random elements are links between stakeholders that do not exist in theory, however, have been identified in reality.

The analysis identified four random elements:

- i. Local communities influence airport surface operations. In theory members of the local community are not in contact with the physical infrastructure of the manoeuvring area or the actors on the manoeuvring area. However, the analysis of safety data revealed circumstances, where members of the local community infringe onto the manoeuvring area. Two examples highlight the issue:
 - a. Children crawling under a gate that was used for construction purposes and infringed onto the airport surface;
 - b. Perimeter fence being snowed under and pedestrians walking over it without realizing the infringement onto the airport and the manoeuvring area.
- ii. Customers (e.g. passengers) are in theory in contact with various service providers (e.g. airline for booking flights, airport operator for information purposes). However, passengers are not allowed onto the airport surface, unless they are sitting in an aircraft. The data revealed that in some circumstances passengers infringed onto the airport surface (e.g. passenger running out to chase lost papers, passenger running out to take pictures of a landing aircraft).
- iii. The airport surrounding land use does not influence the operations on the manoeuvring area. For instance, whether fields, forest, or local communities are established next to the airport does not influence the operations of pilots, V/PDs, or ATC under normal circumstances. However, the



* 3311, 3312, 3313 refer to factors in relation to airport location and surrounding land use, airport landscaping, and manoeuvring area design, respectively.

Fig. 21. Airport surface system architecture.

analysis revealed situations in which this is not true, such as for instance increased bird activity due to a fishing industry next to the airport, which disturbed the operations as increased wildlife removal activities were necessary. Another example includes disruptions caused by birds dragging objects from the surrounding land use onto the manoeuvring area. Examples of objects found in the data include graveyard lights, goofballs, and crab shells.

- iv. Last, the theoretical model shows an interface between the airport surface and its surrounding infrastructure and operations (e.g. apron). Under normal circumstances the operations of apron and surface influence each other. For instance, a pilot starts its operations on the apron and transfers during taxi onto the manoeuvring area. However, the data revealed that there are interfaces between apron and manoeuvring area that are not expected under normal circumstances (i.e. normal operations). These include push-back onto a live taxiway without clearance, operational personnel deliberately disregarding traffic rules and infringing onto taxiways, etc.

In addition, the analysis identified a random interface between the stakeholders. Section 4.1.7 specified communication as the interface between the stakeholders. The safety data analysis revealed a second interface: location, i.e. one actor can influence another party in the same location when infringing onto an area he/she was not supposed to be. Examples include an aircraft infringing onto a runway causing another aircraft to go around, or a vehicle infringing onto a runway and conflicting with an aircraft crossing the landing threshold. The locations of the various stakeholders during normal operations have been described as part of the process model development. Fig. 21 contains the random elements highlighted in red dashed arrows.

4.4. Framework for a holistic risk assessment

Fig. 21 presents the final model of airport surface operations and the key drivers to surface safety. It composes the high-level architecture of surface operations (highlighted in black), its random elements (highlighted in red dashed arrows), and the hazards to the operations as identified through causal factors analysis (represented through blue numbers). The blue numbers represent the relevant parts of the taxonomy as outlined in Section 4.2 Fig. 21.

The model shows the components of the airport surface system architecture and its benefits are threefold. Firstly, a high-level architecture of a system is a necessary prerequisite for SMS development and, therefore, it is recommended as a baseline model for SMS development.

Secondly, the developed model provides a holistic overview of all components/stakeholders involved in surface operations and enables a detailed understanding of the system failures, i.e. factors that cause occurrences, and how these can spread throughout the system. Previous research had failed so far to integrate all relevant components. This has implications for risk mitigation measures, such as the design of training programmes. For instance, while a pilot does not need to understand the causes of failures in the operations of other stakeholders in detail, he/she does need to be aware of them to the extent that they can impact his/her own operations. Training programmes need to account for this in order to allow every stakeholder to develop an understanding about what other actors are doing around them and how that can impact their own operations. In addition, a holistic understanding of operational practices on the airport surface, and associated hazards, improves situational awareness of all stakeholders, which is one of the most frequently cited causes e.g. for incursions (Airbus, 2004; ICAO, 2007). Overall, the developed airport surface system

architecture is applicable to all stakeholders and helps practitioners to think ‘outside the box’, that is, to identify drivers that are not directly related to an operator’s activities, but may indirectly influence their operations and safety. It is recommended as a training and awareness tool and, therefore, it supports the training and education, and safety communication functions, of the safety promotion component of a SMS.

Thirdly, aviation is a fast moving environment and the service providers experience permanent change which requires the adjustment of safety risk controls in the operational environment. The model, therefore, is also recommended as a macroscopic scenario tool for use in SMS change management, thus supporting the safety assurance component of a SMS. It helps to identify how a change to one component can influence the whole system and hence is useful to guide further analysis.

5. Conclusions

In summary, the new model of airport surface operations provides a robust tool for airport surface risk management in the context of SMS. This was achieved through a holistic consideration of surface operations from the viewpoint of all relevant stakeholders and a combination of various rigorous research methods to achieve triangulation. In addition, it was possible to close major gaps in the literature. Its use as (i) baseline for SMS development, (ii) training and awareness tool, and (iii) macroscopic scenario tool to support the SMS change management have been highlighted. Through its robust development the model is applicable for airports in North America, Europe, and Oceania.

The final model allows identifying key drivers to surface safety and helps practitioners to think ‘outside the box’, that is, to identify drivers that are not directly related to an operator’s activities, but may indirectly influence their operations. The model is applicable for airport operators, airlines, ANSP’s, handling companies and regulators and recommended to practitioners for the operational safety management of surface operations in the context of SMS. In addition, the current work is recommended to be used as basis for future research that should focus on the development of more effective safety mitigation strategies, e.g. by quantifying the safety of surface operations from a holistic viewpoint. For instance, using logistic regression models, the functional relationship between airport surface safety occurrences and their underlying causal factors could be modeled. The baseline model as introduced in this paper highlights the components of the airport surface system. Following this model, the impact of each of the system components as well as their interactions on safety (e.g. number of occurrences, severity) should be systematically assessed. This would allow identifying the statistically significant causal factors and help the decision making process for proposing safety recommendations.

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